

Saber Ayatollahi, CFA, MSc

Toronto, Canada • +1 (416) 856-1276 • saber.ayatollahi@gmail.com • www.linkedin.com/in/sayatollahi

SUMMARY

Senior Manager Quantitative Analytics candidate with ~7 years in independent model validation, model governance, and quantitative analytics. Holds delegated sign-off authority at Moody's Analytics on valuation, sensitivity, and ALM model outputs for institutional portfolios of \$5-25bn per engagement, and sits on the model governance committee that helps set benchmarking, documentation, and model-performance standards for client-delivered analytics. Advanced Python (pandas, NumPy, SciPy) and working SQL (PostgreSQL); builds validation tooling and agentic-AI review workflows that cut cycle time 30-40%. CFA charterholder, dual MSc.

CORE SKILLS

Independent Model Validation & Technical Review • Model Governance & Delegated Sign-Off Authority • Conceptual Soundness & Economic-Defensibility Testing • Model Performance Assessment & Benchmarking • Validation Documentation (SR 11-7 / OSFI E-23-aligned) • Python (pandas, NumPy, SciPy) • SQL (PostgreSQL, intermediate) • Stochastic Modelling, Monte Carlo & Scenario Analysis • Derivatives & Fixed Income Valuation Review • Agentic AI Development Workflows (Claude Code, Cursor) • Mentorship & Executive Stakeholder Communication

EXPERIENCE

Moody's Analytics

Asst. Director - Modelling Services

Toronto, May 2023 - Present

Independent Validation & Technical Review

- Holds delegated sign-off authority on valuation, sensitivity, and ALM model outputs for multi-asset institutional portfolios of \$5-25bn per engagement (~\$50bn+ cumulative), independently reviewing curve construction, spread calibration, and cross-asset interactions before production release.
- Tests conceptual soundness and economic defensibility of model assumptions, holding releases and escalating outputs lacking mathematical support to product and development owners for resolution.
- Validates derivatives pricing outputs across rates, FX, and inflation, checking sensitivity consistency and stability at portfolio-level aggregates.
- Reviews aggregation logic converting security-level exposures into portfolio-level risk metrics feeding downstream ALM and capital processes.

Model Governance, Framework & Reporting

- Serves on the model governance committee, contributing to methodology review, benchmarking and documentation standards, and model performance assessment for client-delivered analytics.
- Operates within a formal Model Risk Management and governance framework, building validation and documentation standards consistent with SR 11-7 and OSFI E-23 expectations.
- Acts as escalation contact for client-critical analytical issues and prepares analytical summaries translating quantitative outputs into committee-ready narratives for clients' Heads of Risk.

Quantitative Development, Python & Automation

- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress scenario analysis across ALM horizons.
- Re-engineered manual spreadsheet workflows into scalable Python (pandas, NumPy, SciPy) analytics pipelines with embedded logging, validation, and auditability controls.
- Built agentic AI development workflows (Claude Code, Cursor IDE) automating code generation, validation scaffolding, and anomaly detection, reducing development cycle time an estimated 30-40% on comparable modules.

Client Service Specialist

Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms migrating to the PFaroe DB (risk & ALM) and PFaroe PM analytics platforms, validating model outputs post-deployment.
- Translated client investment-team requirements into structured requests for product owners and development teams, completing migration of all assigned accounts off the legacy Calypso platform.

Ernst & Young (EY)

Senior Consultant - IFRS & Financial Risk Transformation

Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurers, aligning regulatory requirements with system implementation and financial-reporting frameworks.
- Advised actuarial, IT, finance, and program-management functions on standards adoption; contributed to governance documentation and regulatory-readiness processes.

Ortec Finance Canada

Consultant - Asset-Liability Management & Risk

Toronto, Feb 2019 - Sep 2021

- Built and interpreted stochastic economic scenario generators assessing interest-rate risk and funding volatility under base-case and stressed environments.
- Performed asset-only and asset-liability portfolio optimization on VaR and CVaR using Ortec's GLASS platform, with risk decomposition and near-optimal frontier analysis testing the robustness of allocation recommendations.
- Led the ALM study for a three-plan university pension merger on a team of three, developing and validating the model capturing all Funding Policy dynamics across duration, currency, inflation, and leverage overlays.
- Presented study findings and stress-test results to investment committees at Canadian pension-fund clients.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016